

Kosar Amoushahi

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Summary

I'm an AI Engineer at Zharfa Tech and an M.Sc. graduate in Artificial Intelligence from Amirkabir University of Technology (May 2026). Over the past 2 years, I've built production-oriented AI applications, including RAG systems, Persian OCR/VLM document processing, agentic Text-to-SQL, hierarchical document summarization, and document translation workflows—primarily developed with DSPy and FastAPI. Recently I deployed a production RAG stack end to end: benchmarking embedding models on retrieval quality, selecting a model from that evidence, bringing the system up, and tuning retrieval parameters against latency and relevance. I enjoy designing modular, reliable AI systems that integrate LLMs with databases and production services. In parallel, I conducted research in medical imaging; my thesis focused on Heart MRI segmentation using hybrid CNN-Transformer architectures.

Experience

Zharfa Tech — AI Engineer

Feb 2025 – Present

- **Production RAG:** Took embedding selection through deployment and parameter tuning.
 - **Stand up:** Deployed RAGFlow with Docker, with Elasticsearch for chunk search, knowledge bases, and chat assistants, and connected OpenAI-compatible embedding and chat models.
 - **Ingestion and reliability:** Indexed Persian document collections and separated retrieval latency from answer-generation latency so the vector store stayed usable under load. Traced Elasticsearch out-of-memory failures to the memory limit.
 - **Benchmark and model selection:** Built a resumable retrieval benchmark over about 4,000 question-answer pairs. Compared Qwen3-Embedding-4B with jina-v5-small in a retrieve-only setting and scored results with an LLM judge on retrieval quality, so the embedding was chosen from what was retrieved.
 - **Parameter tuning:** Tuned top- n , the similarity threshold, hybrid keyword and semantic search, chunking, and RAPTOR hierarchical indexing, and limited how much context reaches the generator. This cut irrelevant chunks without increasing latency.
 - **Earlier stack:** Brought up a self-hosted R2R system on a Persian corpus of about 2,000 documents and compared vector search with GraphRAG, using Jina embeddings and a resumable indexer that respected API rate limits.
- **Document AI, Persian OCR and RAG ingestion:** Developed an OCR-to-Markdown pipeline for Persian PDF documents using vision-language models, so documents could be ingested for search and question answering.
- **DBQA, database question answering:** Built one system with schema discovery, conversational memory, a DSPy Text-to-SQL agent, and structured answers. The agent writes read-only, schema-safe SQL, runs it on the database, and generates the user-facing answer from the returned rows.
 - When chat history exceeded the context budget, the same system summarized it so the prompt stayed within limits and later SQL and answers stayed consistent.
 - For the accounting database, explored tables and relations, documented the schema and edge cases, and built question-SQL pairs that grounded and improved this agent.
 - Covered the system with unit, integration, and end-to-end tests, plus adversarial prompt checks, to reduce failure modes.
- **Hierarchical summarization agent:** Built a DSPy summarizer with a repository layer for PostgreSQL. It summarizes document chunks hierarchically and writes executive and analytical summaries for different readers.

- **Document translation agent:** Chunked uploaded documents into PostgreSQL, translated each chunk on its own, stored the result, and rebuilt the translated document from those chunks.
- **Fine-tuning vision-language models:** Collected and prepared OCR and document-understanding data, including cleaning, formatting, and quality checks.

Mahya Energy Inc. — Trainee

Jun 2022 – Aug 2022

- Supported MRI segmentation experiments, including data preparation, training runs, and evaluation.

Education

Amirkabir University of Technology, Tehran Polytechnic

Sep 2023 – May 2026

M.Sc. in Artificial Intelligence and Robotics

GPA: 17.48 / 20

Thesis: Heart MRI Segmentation using Hybrid CNN–Transformer Architectures

Isfahan University

Sep 2019 – Sep 2023

B.Sc. in Electrical Engineering

GPA: 17.81 / 20

Skills

LLMs and NLP: RAG, embedding evaluation, LLM-as-a-judge, hybrid search, agentic workflows, prompt engineering, Text-to-SQL, information retrieval

Computer Vision: OCR, vision-language models, image segmentation, classification

Data and Graph: SQL, knowledge graphs, GraphRAG, network analysis

Engineering: service APIs, containers, Linux, version control

Tools & Technologies

DSPy, RAGFlow, Elasticsearch, Python, PyTorch, NumPy, Pandas, Scikit-learn, OpenCV, FastAPI, Docker, Git, GitHub, NetworkX

Courses & Certifications

- Fundamentals of MCP, Hugging Face
- Deep Learning for Images with PyTorch, DataCamp

Teaching Experience

Teaching Assistant — Complex Networks

Amirkabir University of Technology

- Supported students, prepared and maintained assignments, and graded coursework.

Teaching Assistant — Machine Learning

Amirkabir University of Technology

- Supported students, helped with assignments and projects, and graded coursework.

Publications

PHM-TransUNet: A Parallel Hierarchical Multi-Scale Transformer-UNet

- Proposed a parallel hierarchical multi-scale Transformer–UNet for cardiac MRI segmentation, combining MaxViT hierarchical attention, an attention bridge, and multi-scale feature fusion in the decoder. Validated on ACDC with improved Dice.

Honors & Activities

Technology Olympics Competition in Artificial Intelligence

Aug 2024

Speaker, 7th Conference on Cardiology and Emerging Technologies

2025